

Advanced Radio License Notes

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1 Introduction

This are the notes from following the advanced course on HamRadio.au.

2 Electronics and Circuit Theory

2.1 Impedance

2.1.1 Introduction

Impedance is how a circuit stores or resists AC (or any changing) energy. If the impedances within a circuit aren't matched, then power gets reflected back instead of going where it is needed.

Impedance is the combination of resistance and reactance. Resistance is the restriction of the "flow" of energy, and converts that restricted energy to heat. This is measured in Ohms Ω . Reactance can be thought of as a flywheel, it stores energy temporarily and releases it. This is what capacitors and inductors do. This is also measured in Ohms. This means that impedance is the total opposition to current.

Why this matters. The transmitter expects to see 50Ω of impedance. The coax is also 50Ω . If the antenna isn't also close to 50Ω , then the power from the transmitter gets reflected, which is what SWR (standard wave ratio) measures.

2.1.2 Reactance

There are two types of reactance. One is found in inductors, and the other in capacitors.

Inductors (any coil of wire) opposes changes in current, i.e. their reactance increases with frequency

$$X_L = 2\pi fL.$$

This means that the higher the frequency, the more an inductor blocks the signal. This is why they are used in low-pass filters, they all for low frequencies but block high ones.

Capacitors oppose changes in voltage. Their reactance decreases with frequency

$$X_C = \frac{1}{2\pi fC}.$$

The higher the frequency, the easier it passes through the capacitors. This is why a capacitor in series blocks DC but passes RF.

2.1.3 Impedance formula

We can then write impedance as

$$Z = R + jX,$$

where R is the resistance, X is the impedance, and j is used in place of i for imaginary numbers due to i being used for current. The resistance part is where the energy is consumed, and the reactive part is where the energy is stored and returned. The total impedance is then given by the magnitude of this complex number

$$|Z| = \sqrt{R^2 + X^2}$$

Example. An antenna has $R = 50\Omega$ and $X = 25\Omega$. The total impedance magnitude is

$$\begin{aligned} Z &= \sqrt{2500 + 625} \\ &= \sqrt{3125} \\ &\approx 56\Omega \end{aligned}$$

2.2 Resonance

2.2.1 Introduction

When an inductor and capacitor are combined, at one specific frequency their reactances become equal and therefore cancel each other out. This is called resonance, and when this happens the circuit behaves as purely resistive. This is how a single frequency is selected while rejecting everything else.

Radio Example. Tuned circuits within a receiver use this resonance to select a station to hear, and reject everything else. Antennas are resonant at a specific frequency, which is where it works best.

The formula to find the resonant frequency is given by

$$f_r = \frac{1}{2\pi\sqrt{LC}}.$$

This says that for a given inductor L and capacitor C , there's exactly one frequency where they resonate. This is an important formula.

2.2.2 Series and Parallel Resonance

Series and parallel arrangements of L and C result in opposite behaviour at resonance.

Table 1: Behaviour of circuits at resonance		
	Series LC	Parallel LC
Impedance	Minimum (just R)	Maximum
Current	Maximum	Minimum from source
Used for	Passing a desired frequency (band-pass)	Blocking a desired frequency (or selecting it) (band-stop)

Radio Example. A series-resonant circuit in an antenna trap passed the trap's frequency through, while a parallel-resonant circuit in a receiver's IF (intermediate frequency) stage has high-impedance at the desired frequency, which develops a large signal voltage across it.

2.2.3 Q-Factor

The Q-factor described how selective a resonant circuit is. A higher Q means a very sharp tuning (with narrow bandwidth), and low Q means a very broad tuning (wide bandwidth)

$$Q = \frac{f_r}{\text{Bandwidth}}$$

For a series RLC , $Q = \frac{X_L}{R}$, so less resistance means higher Q.

Bandwidth Rule. Bandwidth is $\frac{f_r}{Q}$, so a circuit with $Q = 100$ at 7 MHz has a bandwidth of 70 kHz, which is enough to cover a portion of the 40m band. A crystal filter with $Q = 50000$ at 9 MHz has a bandwidth of only 180 Hz, which is perfect for CW.

2.3 Filters

2.3.1 Introduction

Filters are extremely important in radio. They are what selects the frequencies that get let through and what get blocked. Without filters, a receiver would hear literally everything at the same time, and the transmitter would spray harmonics all over the spectrum (which happens to be illegal).

2.3.2 Types of Filter

There are four basic types of filter.

Table 2: Basic types of filter		
Filter Type	Behaviour	Where it is in radios
Low-pass	Passes low frequencies, blocks high frequencies	After the transmitter PA (power amplifier). It blocks harmonics before reaching the antenna
High-pass	Passes high frequencies, block low frequencies	At a TV antenna input. It blocks HF signals, but lets VHF/UHF TV through
Band-pass	Passes a specific range, blocks everything else	IF (intermediate frequency) filters in a receiver. It selects the signal bandwidth that you want.
Band-stop (notch)	Blocks a specific range, passes everything else	Notch filter to remove an interfering carrier.

2.3.3 LC Filter

Filters use inductors and capacitors in combinations to make a specific filter. From above, we know

- Inductors block high frequencies (reactance increases with frequency)
- Capacitors block low frequencies (reactance decreases with frequency)

Low-pass Filter. A low-pass filter wants to block high frequencies, and allow low frequencies through. This means that the circuit would put an inductor in series (to block high frequencies), and capacitors to ground (short high frequencies to ground).

2.3.4 Filter Steepness

In practice, filters can't instantly go from passing a frequency to blocking another. The change is gradual, and this is affected by the number of LC sections (called a pole). Each pole adds 20 dB/decade of rolloff. The higher this number, the more steep the dropoff, as it is saying that per decade (1 decade is a ratio of 10 between two numbers on a logarithmic scale), the scale drops by 20 dB.

Example. Consider a transmitter that produces a 7 MHz signal. The 2nd harmonic at 14 MHz and 3rd harmonic at 21 MHz must be suppressed. A good 5-element low-pass filter with a cutoff just about 7 MHz can suppress the 2nd harmonic by 40+ dB and the 3rd by 60+ dB. This is the difference between a clean signal and an interference complaint.

A typical transmitter low-pass filter has 5-poles (100 dB/decade, a steep slope). For critical applications, a 7-pole filter is used (140 dB/decade, a very steep slope).

2.3.5 Crystal Filters

Crystal filters have very high Q factors (10 000 to 100 000+). Multiple crystals combined in a so-called ladder filter give the sharp selectivity for IF stages

- 2.4 kHz bandwidth - for SSB reception
- 500 Hz bandwidth - for CW reception
- 6 kHz bandwidth - for AM reception

In modern radios however, crystals are being replaced with DSP (digital signal processing) filters, as they allow changing the bandwidth on the fly.